CHAPTER 3 km reanalysis — GRIB2 archive contents

90 variables, 246 GRIB messages per hourly file. Taken from the converter's own registry (`wrf_era5_comparison.WRF_TO_ECMWF_PARAMID`), which is what the pipeline and its sanity check use, so this document describes what is actually written. Document updated on 2026-09-18, when the WPS static file `geo_em.d02` was recovered from LRZ after having been recorded as lost. It carries the FRACTIONAL land cover, soil texture and lake data that the hourly output had reduced to a single dominant category per cell, and it added fifteen fields: `cvl`, `cvh`, `tvl`, `tvh`, `slt`, `cl`, `dl` and the eight ERA5 soil layers. Nothing was removed. Every field here is one we can defend row by row. Where the WRF quantity is not exactly the ERA5 one, it is said so below rather than hidden; where it could not be made to correspond at all, the field was dropped instead of published under a name that would mislead (see §6). This document says what is in the files. Two companions say the rest: `MISSING_VARIABLES.md` says what is not and why, and `CHAPTER_KNOWN_PROPERTIES.md` states the **known properties and biases** of what is — the places where the archive is correct and surprising, each with the measured number. Read the third one before comparing this archive against ERA5 or against observations.

1. What the files are

Model	**WRF 4.1.1, domain d02, 49 vertical levels, model top 50 hPa**
Horizontal grid	Mercator, **1641 × 1353** points, **3 km** spacing (2 220 273 points/field)
Domain corners	23.472 N, 19.972 W → 60.001 N, 42.256 E; standard parallel 44.671 N
Earth shape	**Sphere of radius 6 370 000 m** (`shapeOfTheEarth=1`) — WRF's own sphere, declared explicitly
GRIB edition	**GRIB2**, grid definition template **3.10** (Mercator), originating centre `ecmf`
Packing	`grid_ccsds`, **24 bits per value**; masked fields carry a proper GRIB bitmap
Parameter identity	**ECMWF paramId / shortName** throughout, so the archive is directly comparable with ERA5
File	one file per hour, **~737-765 MB**: measured means of 737 MB over 184 files of September 2024 (661 MB at 00Z) and 765 MB over 476 files of February-March 2024. That is ~17.7-18.4 GB/day and **~6.5-6.7 TB/year**
Naming	`ailam-an-cima-3km-{year}-{year}-1h-v1-{YYYYMMDD}{HH}.grib`
Period	2019, 2024 and 2025-01-01 → **2025-07-02**, hourly (2020–2023 to follow). The 2025 end is where the WRF run itself stops: an `mmlsattr` scan of the whole of 2025 on the source (2026-09-29) finds one contiguous block of 4392 hours ending 2025-07-02T23 and nothing after it

2. Vertical levels

13 pressure levels, the same for every 3-D variable:

1000, 925, 850, 700, 600, 500, 400, 300, 250, 200, 150, 100, 50 hPa

They are the ERA5 set truncated at the model top: the 8 ERA5 levels above 50 hPa are above WRF's `P_TOP` and cannot be produced. Single-level fields sit on `surface`, `heightAboveGround` (2, 10, 100 and 200 m), `meanSea`, `entireAtmosphere` (column integrals) or, for the radiation budget at the top of the atmosphere, `nominalTop`.

3. Time convention — read this before indexing the archive

- **Instantaneous fields** use product definition template **0**: `dataDate`/`dataTime` are the valid time.
- **Accumulated fields** (`tp`, `tirf`, `sf`, `sro`, `ssro`, `ro`, and the whole radiation budget) use template **8**, as their ECMWF definitions require, and each covers **the hour ending at the validity time**, as ERA5 publishes them: reference time H−1, step `0–1`. The 00 UTC file holds the hour 23→00 UTC of the previous day. A day's total is the sum of its 24 hourly values, 01 UTC to 00 UTC of the next day.
- **At 00 UTC the accumulations and the instantaneous fields come from different model runs.** Each run starts at 18 UTC of the day before the one it covers and ends at 00 UTC of the day after; the 00 UTC file's instantaneous fields come from the run for its own day, and its accumulations from the run for the day before, the only one that integrated that hour.
- **`10fg`** is a **maximum over the preceding hour** (reference time H−1, step `0–1`), like the accumulations, and it is the maximum of the *resolved* 10 m wind, not a gust parameterisation.
- A file therefore mixes reference times on purpose. **Index on validity (`validityDate`/`validityTime`), never on `dataTime`.**
- To tell an accumulation from an instantaneous field, test `stepType` (or `typeOfStatisticalProcessing`) and `stepRange`. **Do not use `generatingProcessIdentifier`**: it is 128 in every one of the 246 messages, because 128 is eccodes' own GRIB2 default, so inside this archive it discriminates nothing.

4. Declared approximations

Six fields, in five rows, are not exactly their ERA5 namesake, and are published with that said:

field	what ERA5 means	what we write
`slor`	slope of the **sub-grid** orography, a parameter of the form-drag scheme	the slope of the **resolved** 3 km terrain, `|grad z|`. Note eccodes will show you the parameter's own name, *Slope of sub-gridscale orography*, so nothing in the message itself says this. `sdor` beside it **is** the ERA5 quantity, only evaluated on a 3 km cell instead of a 31 km one: the measured ratio of medians is 0.244, which is (3/31)^0.605 — a resolution rescaling, not a different definition
`10fg`	gust from a parameterisation (turbulent + convective)	the hourly maximum of the **resolved** 10 m wind
`iews`, `inss`	the model's own stress components	the magnitude `rho u*^2`, projected on the 10 m wind direction

field	what ERA5 means	what we write
`fsr`	an **effective** roughness, carrying the orographic drag the IFS parameterises at 31 km	WRF's **local** `ZNT`: the land-use roughness with a snow override. Over land it runs at **0.32–0.48 of ERA5's** in the median; over sea the two agree, both being Charnock. `zust` beside it is the model's own friction velocity and runs **1.6–2.1× ERA5's over land** (1.1× over sea) — the other half of the same story, because at 3 km the terrain-driven wind is resolved rather than smoothed
`lsm`	a **permanent** land-sea mask; sea ice is carried separately in `ci`	WRF's `LANDMASK`, which reclassifies sea-ice points as land, so the mask **grows in winter**. `lsm & ~(ci > 0)` recovers the ERA5 sense exactly, and `ci` is published

`lsm` is now the **only** field that follows the model hour by hour (with `al`, which moves with it). The soil columns used to as well, and no longer do: under new sea ice the scheme grows a column that is not soil — `swvl` reached exactly 1.000 against its own soil type's porosity of 0.435 — so since 2026-09-25 they are masked on the **permanent** land of the static file, as ERA5 does. Nothing is lost that cannot be recovered: `ci` is published. Six more deserve a note rather than a warning:

- **`skt`** is not a model output here (`TSK` was not written): it is inverted from the upward longwave, `LWUPB = eps sigma T^4 + (1-eps) LWDN`. The emissivity is not assumed — it was **read out of the run itself**. The radiation scheme computes a clear-sky diagnostic with the same emissivity and the same skin temperature, changing only the downward flux, so the two equations together eliminate the unknown temperature and leave `eps = 1 - (LWUPB - LWUPBC)/(LWDN - LWDNC)`. That is an identity, not a fit. Measured this way the emissivity came out identical in every cell of a given land-use category (spread 0.0000 to four decimals), and the control is exact: over open water, where the model's skin temperature IS the sea surface temperature, it returns 0.98000.

The accuracy is seasonal, and both numbers matter. The reconstruction reproduces the model's own emissivity to 1e-4 on **99.98 % of land points in July and 93.05 % in January**, for a land `skt` error of **0.0012 K RMSE in July against 0.084–0.119 K in January and February** (p99 0.81 K, worst point 2.79 K). All of that spread is the snow rule: snow raises the emissivity to a flat 0.98 in every category, applied from 1.5 % snow cover upwards, with a blend below it. That threshold is the only quantity in this derivation that was chosen rather than read out of the run, so it was measured: swept over all 34 sample timesteps against the emissivity the model actually used, 1.5 % is the minimum of the resulting skin-temperature error. Over open water, where the answer is known independently, it recovers the sea surface temperature to **0.0005 K RMSE** on about 910 000 points at every timestep of every season. Note this derivation error, 0.119 K RMSE at its worst (February), is more than ten times smaller than the physical cold bias of the land skin itself — which is the separate and much larger fact: **`skt` over land runs about 1.6 K below ERA5, reaching 3.2 K at midday**, while over sea it is exact (+0.01 K). That is the run, not the derivation; the sea number is the control that proves it. Reconciling it against the extra shortwave and the missing cloud above would need the surface energy budget, and `HFX`, `LH` and the ground heat flux are the three fields this archive does not have.

- **`al`** is a two-season table value, not a field. No monthly satellite albedo was used (`usemonalb =`

`.false.`), so the snow-free background comes from the land-use table's WINTER and SUMMER columns by dominant category: it takes only **8 to 11 distinct values** over the whole domain, and it changes at **1 171 937 land points — 89.9 % of the land mask — at the seasonal boundary**, which is April and November, not a physical transition. Off land it is 0.08 on water and 0.65 on sea ice. Do not read it as an observed or evolving albedo, and do not difference it across the boundary expecting a signal. `fal`, beside it, is the actual ratio the simulation used and does evolve.

- **The surface shortwave carries a systematic high bias.** RRTMG ran with an ozone climatology and **no aerosol at all** (`aer_opt = 0`), while ERA5 carries the Tegen aerosol climatology; over this domain an optical depth of 0.1 to 0.2 accounts for the size. Measured against ERA5: **+8.5 % under clear sky** (`ssrdc`) and **+13.6 % all-sky** (`ssrd`). The thermal fields are untouched by it and agree to within 1.1 %, which is the control that makes the attribution credible. Total cloud cover also runs **22 % below** ERA5 and high cloud **39 % below**, which pushes surface solar the same way. Both are properties of the run and cannot be removed after the fact; they and the rest of the archive's measured biases are stated in full in `CHAPTER_KNOWN_PROPERTIES.md`.
- **Below the ground the pressure levels are clamped, not extrapolated.** Where a level lies under the terrain, the archive repeats the **lowest model level's value verbatim** — matched to exactly 0.0 for `t`, 1e-9 for `q`. That is a real model state from about 24 to 27 m above the ground rather than synthetic air, and it keeps the surface join coherent. But **ERA5 extrapolates there with a lapse rate**, so the two archives differ in convention exactly where 1000 hPa is underground, which is most of the land; and our `z` at 1000 hPa then sits a median of 24 to 27 m *above* the terrain that `sp` puts it below, on **47.6 % of the domain** at 1000 hPa and 9 % at 925. Mask the levels against `sp` if you need a clean comparison — every field needed to do it is in the same file.
- **`sst`** is constant within each 24 h run (`SST_UPDATE=0`), re-initialised daily.
- **`dl`** is the lake depth database the run was built with, not a model state: lake physics was off (`sf_lake_physics=0`), so nothing in the simulation responds to it. It is published as the boundary dataset it is, and masked off-lake — the raw field is the 10 m WPS default fill on 99.56 % of the domain, and 79 464 of the 88 785 lake cells carry that same default.

5. Masked fields

Written with a GRIB bitmap rather than fake zeros, so missing means missing:

field	defined where
`sst`, `ci`	**sea** points only — and sea here is not simply "not land": WPS's land-use class calls the Black Sea, the Sea of Azov and part of the Baltic a *lake*, so masking on that class would have deleted `sst` from the Black Sea. The discriminator is connected-component size, measured: the water bodies of this domain are 844 431 cells (Atlantic + Mediterranean + Baltic + North Sea), 49 841 (Black Sea + Azov), 8 583 (Red Sea), then 1 225 (Vänern, the largest genuine lake) and smaller, so the cut sits in a gap with a factor of seven of slack
`cl`	zero on the sea rather than missing, since a sea point genuinely has no lake cover

field	defined where
`mucin`	the columns where the CAPE routine returns a value at all, i.e. where CAPE reaches 100 J/kg. **That is 3.8 % (January) to 28.8 % (July) of the domain**. The gaps are written as missing rather than zero, because zero would read as ”nothing inhibits convection” at precisely the columns where the most does
`tsn`	cells whose snow cover exceeds 0.9 (≈ 29 mm water equivalent); out of season the field is legitimately empty
`fal`	daytime only (incoming shortwave above 50 W/m²)
`swvl1-4`, `stl1-4`	the **permanent** land of the static file — a fixed outline, the same in every file of the archive. Sea ice therefore carries no soil column, which is what ERA5 does too
`slt`	land points of the static file (1 304 109 cells)
`tvl`	everywhere except the cells where a shrubland dominates the low vegetation (124 672 cells, 16.02 % of the 778 381 with `cvl > 0`). MODIS carries no shrub phenology while ECMWF’s table splits evergreen from deciduous shrubs, and rather than invent one the field is left missing. `cvl` still gives the cover there
`dl`	cells with a lake cover above zero (32 000)

6. What is not in the archive

No TKE and no boundary-layer height (the PBL scheme is YSU, non-local); no ocean waves, currents, sea surface height or sea-ice thickness (no ocean or wave coupling); no convective precipitation (`CU_PHYSICS=0`, so it is identically zero and would be a dead field). Deliberately dropped rather than approximated:

- **leaf area index** (`lai_lv`, `lai_hv`): the model carries a single LAI per cell, while ERA5 wants one per vegetation class, defined per unit of vegetated area. Any split between the two would be our assumption, so neither field is written. **The single field itself is in the model output** — alive, and unpublished for that reason alone; it can be delivered under its own name, `LAI`, if it is wanted. Note `rdlai2d = .false.`, so it is a table lookup on the dominant category rather than an observation, which is a second reason not to split it;
- **sub-grid orography angle and anisotropy** (`anor`, `isor`): the static file carries the Kim-Arakawa asymmetry and effective-length parameters, which are not ECMWF’s, and the difference does not fit in one line.

Four entries that stood here until 2026-09-17 have moved into the archive, because `geo_em.d02` came back and because the soil objection had an answer:

- **`swvl1-4` / `stl1-4`** are now the true ERA5 layer averages. The land-surface scheme is a *level* scheme — point values at 0, 5, 20, 40, 160 and 300 cm with a linear profile between them — and the whole ERA5 column lies inside those nodes (289 cm is above the deepest, 0 cm is the first). So each layer average is the exact integral of the model’s own profile and **nothing is extrapolated**. The objection was against writing the point values, not against integrating them;

- **`cvl` / `cvh`** are sums of the fractional cover over the MODIS-IGBP classes that are low and high vegetation — the same quantity ERA5 means. This is what the dominant category could not express: it holds only 90.1 % of a land cell on average, and 17.9 % of land cells carry both high and low vegetation above 5 %;
- **`tvl` / `tvh`** are the dominant low and high class translated to ECMWF code table 4.234 by the table in §8, where each row is a semantic identity. The one place MODIS cannot answer the question — evergreen versus deciduous shrubs — is left missing rather than guessed;
- **`slt`** is computed with the FAO clay/sand thresholds that *define* ECMWF's seven soil types, applied to the static file's own clay and sand fractions. No category crosswalk is involved, so the quantity is ERA5's by construction. One caveat for a strict WMO reader: eccodes resolves `slt` to discipline 2/3/0, which WMO table 4.213 reads as a different 11-value list. This archive is ECMWF semantics throughout, and so is this message.

`cl` and `dl` came with them. `cl` is not redundant with the land-sea mask: 19 309 cells that the mask calls land carry a sub-grid lake, and the model itself discarded the class at run-time. The full reasoning, and the routes by which some of it could still be obtained, is in `MISSING_VARIABLES.pdf`.

7. The variables

Descriptions and units are the ECMWF ones. "Time" says how the field is sampled. There are 90 rows and 89 distinct shortNames, because `z` is written twice: as geopotential on the pressure levels and as the surface geopotential (orography). variables=90 messages=246

Pressure levels (13) — 13 variables, 169 messages

shortName	paramId	Description	Units	Level	Time
cc	248	Fraction of cloud cover	(0-1)	13 levels	instantaneous
ciwc	247	Specific cloud ice water content	kg/kg	13 levels	instantaneous
clwc	246	Specific cloud liquid water content	kg/kg	13 levels	instantaneous
crwc	75	Specific rain water content	kg/kg	13 levels	instantaneous
cswc	76	Specific snow water content	kg/kg	13 levels	instantaneous
q	133	Specific humidity	kg/kg	13 levels	instantaneous
r	157	Relative humidity	%	13 levels	instantaneous
t	130	Temperature	K	13 levels	instantaneous
u	131	U component of wind	m/s	13 levels	instantaneous
v	132	V component of wind	m/s	13 levels	instantaneous
w	135	Vertical velocity (pressure)	Pa/s	13 levels	instantaneous

shortName	paramId	Description	Units	Level	Time
wz	260238	Geometric vertical velocity	m/s	13 levels	instantaneous
z	129	Geopotential	m^2/s^2	13 levels	instantaneous

Height above ground — 11 variables, 11 messages

shortName	paramId	Description	Units	Level	Time
100u	228246	100m U component of wind	m/s	100 m	instantaneous
100v	228247	100m V component of wind	m/s	100 m	instantaneous
10fg	49	10m wind gust (hourly max of resolved wind)	m/s	10 m	max over the previous hour
10u	165	10m U component of wind	m/s	10 m	instantaneous
10v	166	10m V component of wind	m/s	10 m	instantaneous
200u	228239	200m U component of wind	m/s	200 m	instantaneous
200v	228240	200m V component of wind	m/s	200 m	instantaneous
2d	168	2m dewpoint temperature	K	2 m	instantaneous
2r	260242	2m relative humidity	%	2 m	instantaneous
2t	167	2m temperature	K	2 m	instantaneous
vwsh	260068	Vertical speed shear (bulk 10-100 m)	s^-1	100 m	instantaneous

Mean sea level — 1 variables, 1 messages

shortName	paramId	Description	Units	Level	Time
msl	151	Mean sea level pressure	Pa	mean sea level	instantaneous

Total column — 10 variables, 10 messages

shortName	paramId	Description	Units	Level	Time
hcc	188	High cloud cover	(0-1)	column	instantaneous

shortName	paramId	Description	Units	Level	Time
lcc	186	Low cloud cover	(0-1)	column	instantaneous
mcc	187	Medium cloud cover	(0-1)	column	instantaneous
tcc	164	Total cloud cover	(0-1)	column	instantaneous
tciw	79	Total column cloud ice water	kg/m^2	column	instantaneous
tclw	78	Total column cloud liquid water	kg/m^2	column	instantaneous
tcrw	228089	Total column rain water	kg/m^2	column	instantaneous
tcsw	228090	Total column snow water	kg/m^2	column	instantaneous
tcw	136	Total column water	kg/m^2	column	instantaneous
tcwv	137	Total column water vapour	kg/m^2	column	instantaneous

Surface / single level — 47 variables, 47 messages

shortName	paramId	Description	Units	Level	Time
al	174	Albedo (climatological, snow-free background)	(0-1)	surface	instantaneous
ci	31	Sea ice area fraction	(0-1)	surface	instantaneous
cl	26	Lake cover	(0-1)	surface	instantaneous, static
cvh	28	High vegetation cover	(0-1)	surface	instantaneous, static
cvl	27	Low vegetation cover	(0-1)	surface	instantaneous, static
fal	243	Forecast albedo (upward/downward SW at surface)	(0-1)	surface	instantaneous
fsr	244	Forecast surface roughness	m	surface	instantaneous
iews	229	Instantaneous eastward turbulent surface stress	N/m^2	surface	instantaneous

shortName	paramId	Description	Units	Level	Time
inss	230	Instantaneous northward turbulent surface stress	N/m^2	surface	instantaneous
lsm	172	Land-sea mask	(0-1)	surface	instantaneous
ro	205	Runoff (surface + sub-surface)	m	surface	hour ending at validity
rsn	33	Snow density	kg/m^3	surface	instantaneous
sd	141	Snow depth (water equivalent)	m	surface	instantaneous
sdor	160	Standard deviation of orography	m	surface	instantaneous
sf	144	Snowfall (water equivalent)	m	surface	hour ending at validity
skt	235	Skin temperature (inverted from upward longwave)	K	surface	instantaneous
slor	163	Slope of orography (resolved, 3 km)	Numeric	surface	instantaneous
slt	43	Soil type (ECMWF 7 classes)	~	surface	instantaneous, static
snowc	260038	Snow cover	%	surface	instantaneous
sp	134	Surface pressure	Pa	surface	instantaneous
src	198	Skin reservoir content	m	surface	instantaneous
sro	8	Surface runoff	m	surface	hour ending at validity
ssr	176	Surface net solar radiation	J/m^2	surface	hour ending at validity
ssrc	210	Surface net solar radiation, clear sky	J/m^2	surface	hour ending at validity
ssrd	169	Surface solar radiation downwards	J/m^2	surface	hour ending at validity
ssrdc	228129	Surface solar radiation downwards, clear sky	J/m^2	surface	hour ending at validity
ssro	9	Sub-surface runoff	m	surface	hour ending at validity
sst	34	Sea surface temperature	K	surface	instantaneous

shortName	paramId	Description	Units	Level	Time
stl1	139	Soil temperature level 1 (0-7 cm)	K	surface	instantaneous
stl2	170	Soil temperature level 2 (7-28 cm)	K	surface	instantaneous
stl3	183	Soil temperature level 3 (28-100 cm)	K	surface	instantaneous
stl4	236	Soil temperature level 4 (100-289 cm)	K	surface	instantaneous
str	177	Surface net thermal radiation	J/m^2	surface	hour ending at validity
strc	211	Surface net thermal radiation, clear sky	J/m^2	surface	hour ending at validity
strd	175	Surface thermal radiation downwards	J/m^2	surface	hour ending at validity
strdc	228130	Surface thermal radiation downwards, clear sky	J/m^2	surface	hour ending at validity
swvl1	39	Volumetric soil water layer 1 (0-7 cm)	m^3/m^3	surface	instantaneous
swvl2	40	Volumetric soil water layer 2 (7-28 cm)	m^3/m^3	surface	instantaneous
swvl3	41	Volumetric soil water layer 3 (28-100 cm)	m^3/m^3	surface	instantaneous
swvl4	42	Volumetric soil water layer 4 (100-289 cm)	m^3/m^3	surface	instantaneous
tirf	235015	Time integral of rain flux (rain only)	kg/m^2	surface	hour ending at validity
tp	228	Total precipitation	m	surface	hour ending at validity
tsn	238	Temperature of snow layer	K	surface	instantaneous

shortName	paramId	Description	Units	Level	Time
tvh	30	Type of high vegetation (ECMWF code table 4.234)	~	surface	instantaneous, static
tvl	29	Type of low vegetation (ECMWF code table 4.234)	~	surface	instantaneous, static
z	129	Geopotential (surface)	m^2/s^2	surface	instantaneous
zust	228003	Friction velocity	m/s	surface	instantaneous

”static” in the Time column means the field is identical in every file of the archive: it comes from the run’s static input, not from the hourly state. `lsm` is deliberately NOT static — the model reclassifies sea points as land where sea ice appears, so it changes with the season.

Entire lake — 1 variable, 1 message

shortName	paramId	Description	Units	Level	Time
dl	228007	Lake total depth	m	entireLake	instantaneous, static

Top of atmosphere / most unstable parcel — 7 variables, 7 messages

shortName	paramId	Description	Units	Level	Time
mucape	228235	Most-unstable CAPE	J/kg	most unstable parcel	instantaneous
mucin	228236	Most-unstable CIN	J/kg	most unstable parcel	instantaneous
tisr	212	TOA incident solar radiation	J/m^2	top of atmosphere	hour ending at validity
tsr	178	Top net solar radiation	J/m^2	top of atmosphere	hour ending at validity
tsrc	208	Top net solar radiation, clear sky	J/m^2	top of atmosphere	hour ending at validity
ttr	179	Top net thermal radiation	J/m^2	top of atmosphere	hour ending at validity
ttrc	209	Top net thermal radiation, clear sky	J/m^2	top of atmosphere	hour ending at validity

8. How the land-surface categories were translated

`tvl`, `tvh` and `slt` are the only fields in the archive that carry a *code* rather than a number, so the tables that produced them are published here: without them the codes cannot be read, and the mapping cannot be checked. The source is the run’s own static file `geo_em.d02` (MODIS-IGBP 21 classes, WPS `MMINLU=MODIFIED_IGBP_MODIS_NOAH`). Its land use was verified against the hourly

output: it agrees with the model's dominant category on **100.00 %** of points (July 2019, January 2025) once one expected difference is excluded — the model recodes lake to water at runtime, because lake physics was off. With sea ice present the agreement is 99.98 % (2025-02-15 12Z), and every one of the 2735 differing points is a sea-ice point, reclassified by the model. This is what makes `cl` recoverable at all.

`tvl` / `tvh`: MODIS-IGBP → ECMWF code table 4.234

High and low follow the IGBP class definitions of tree cover, not a canopy-height table.

MODIS-IGBP class	**set**	**ECMWF 4.234**
1 Evergreen Needleleaf Forest	high	3 Evergreen needleleaf trees
2 Evergreen Broadleaf Forest	high	6 Evergreen broadleaf trees
3 Deciduous Needleleaf Forest	high	4 Deciduous needleleaf trees
4 Deciduous Broadleaf Forest	high	5 Deciduous broadleaf trees
5 Mixed Forests	high	18 Mixed forest/woodland
8 Woody Savannas	high	19 Interrupted forest (both are 30–60 % tree cover)
9 Savannas	low	7 Tall grass
10 Grasslands	low	2 Short grass
11 Permanent Wetlands	low	13 Bogs and marshes (absent from this domain)
12 Croplands	low	1 Crops, mixed farming
14 Cropland/Natural Mosaic	low	1 Crops, mixed farming (absent from this domain)
18 Wooded Tundra, 19 Mixed Tundra, 20 Barren Tundra	low	9 Tundra
6 Closed Shrublands, 7 Open Shrublands	low	**not translated — `tvl` is missing there**
13 Urban, 15 Snow and Ice, 16 Barren, 17 Water, 21 Lake	neither	— (not vegetation; they are the remainder of `1 − cvl − cvh`)

Every translated row is a semantic identity: the ECMWF type names the same vegetation as the MODIS class. The two shrubland classes are the exception and are the reason `tvl` carries a mask. MODIS does not record whether a shrubland is evergreen or deciduous, while table 4.234 has separate codes for the two (16 and 17). Choosing one would have been our invention over 16.02 % of the cells with low vegetation (124 672 of 778 381), so the field is left missing there instead; `cvl` still reports the cover. The classes that would have been the other ambiguous rows, 11 and 14, do not occur in this domain at all.

`slt`: computed, not translated

No category crosswalk is involved. ECMWF's seven soil types are *defined* by clay and sand percentage thresholds, and the static file carries clay and sand fractions, so the definition is applied directly:

condition (percent)	**ECMWF soil type**
clay ≥ 60	5 very fine
35 ≤ clay < 60	4 fine
clay < 35 and sand ≤ 15	3 medium fine

condition (percent)	ECMWF soil type
(18 ≤ clay < 35 and sand > 15) or (clay < 18 and 15 < sand ≤ 65)	2 medium
clay < 18 and sand > 65	1 coarse
dominant soil category is *organic material*	6 organic

Every land point classifies; type 7 (tropical organic) does not occur here. Distribution over the 1 304 109 land points: 1 069 456 medium, 187 378 coarse, 45 450 fine, 1 498 organic, 327 very fine.

`swvl1-4` / `stl1-4`: integrated, not relabelled

The land-surface scheme is a *level* scheme: it carries point values at 0, 5, 20, 40, 160 and 300 cm, with a linear profile between them. ERA5 wants averages over 0–7, 7–28, 28–100 and 100–289 cm. Those four intervals lie entirely inside the model's own nodes — 289 cm is above the deepest one and 0 cm is the first — so each ERA5 layer is the **exact integral** of the model's profile over it, and nothing is extrapolated at either end. In practice this is a fixed 4 × 6 matrix of weights, applied to the six model levels. `swvl` uses the total soil moisture (liquid plus ice), which is what the ERA5 name means; the liquid-only field is not used. Both sets are masked off the **permanent** land of the static file, so over water and over sea ice alike.